



United States International University

Course Outline: 2026 FALL

SWE4040A Software Construction and Development

Lecturer: Wangai Mambo **Venue:** SS16 **Time:** 7:20-9:00 PM **Days:** Tue-Thurs

Email: wnmambo@usiu.ac.ke **Dates:** 7/09/2026

Course Description

This will ensure that students are able to develop large real life usable software with thousands of lines of code.

Credit: 4.5 Credits

Prerequisite: SWE3050

CLO	Aligned to the following university mission outcomes:				
	Higher order thinking	Global understanding and	Community service	Preparedness for career	Leadership and Ethics
1. Demonstrate understanding of the concepts of innovation, growth and value creation	✓	✓		✓	✓
2. Describe the processes by which IT innovation is fostered, managed, and commercialized	✓	✓		✓	✓
3. Spot new IT business opportunities in the environment, whether by recognition, development, or creation	✓	✓		✓	✓
4. Effectively and efficiently evaluate the potential of new IT business opportunities	✓	✓		✓	✓
5. Assess the market potential for a new venture, including customer need, competitors, and industry attractiveness	✓	✓		✓	✓
6. Understand the key components of a successful business model and be able to pitch a venture to a potential funder or customer	✓	✓		✓	✓

Link to Program Learning Outcomes

PLO	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6
	Demonstrate the use of IS in a business organization	Evaluate IT problems in a business organizations .	Apply communication & collaboration skills in IT projects.	Design IT solutions for business organizations.	Solve IT problems in business organizations	Manage IS within their area of specialization.
APT 2080 INTRODUCTION TO SOFTWARE ENGINEERING	A	A	A	A	A	A

Expected Learning Outcomes

By the end of this course, students will:

1. Understand software construction concepts and principles for large software systems
2. Understand nature of software construction problems
3. Improving individual and group software construction process
4. Strategize on how prepare for software construction futures based on emerging trends
5. Apply software engineering methods to develop large software systems

Teaching Methodology

A series of lectures, group discussions. Audio-visual aids will be used in the lectures. Students will make frequent use of Case Studies in understanding the concepts and practices of software engineering, software development mini projects

Information literacy intensiveness

Students will be given software engineering assignments where they will be expected to formulate explicit search strategies and evaluate search results for quality and reliability. Two second half of semester assignments and Final project will be based and assessed on writing and information literacy intensive criteria.

Instructional Material &/or Equipment

Textbooks, whiteboard, handouts, electronic projector, Internet access, and library.

Method of Evaluation

Class/Lab assignments, project work, Research & Case Presentations, written exams, will be used to evaluate the students;

Attendance	3%
Assignments and Exercise	17%
CATS	8%
Projects	22%
Mid-semester	20%
Final semester exam	30%

Total

100%**Content & Class Schedule**

Week	Topics	Reading, exercises and CATS
1	Practical software construction and development Concepts and principles of software engineering Software construction trends <u>Individual Lesson exercise1 during class submit</u>	
2	<u>Large software systems and Software practice</u> <u>Putting XP into practice</u> <u>Fundamentals of software construction</u>	
3	Issues in software design Software structure and architecture	
4	Constructing software with agile methods Scaling agile methods Developing a software system - case study	CAT1
5	Software design methods and strategies	Developing a book recommender software systems
6	Teaming and leadership Spotify + Erickson case studies	
7	Mid Semester Exam	Revision
8	Software intensive systems Planned based methods - Spiral software development method	
9	Individual and team engineer capability building - <u>Personal</u> software process, <u>team</u> and people capability building Spiral software development method	
10	Future of software construction and emerging trends	CAT2 (to be decided)
11	Group project	Work on projects
12	Group project	Work on projects

	Submit group project draft	
13	Group project Submit final complete group/individual project	Finish projects and submit
14	Final Comprehensive Exam	Revision

Core Reading Materials

1. Torgeir Dingsøy, T. Dybø, T. Moe, N. B. (2010). Agile Software Development Current Research and Future Directions. Springer.

Recommended Reference Materials

2. Sommerville, I. (2016). Software Engineering 10th Edition, Pearson
3. Pfleeger Lawrence Shari (2010), Software engineering theory and practice, Pearson.
4. Pressman, R. S. (2010) Software Engineering, A Practitioner's Approach, 6th Edition, McGraw-Hill.
5. Hunt, J. (2006). Agile Software Construction, Springer.
6. USIU Library Software Engineering Books

Journals

1. Wiley Software: Practice and Experience Journal
2. The Journal of Object Technology
3. IEEE Software
4. Software and Information Technology
5. Software Journal
6. IEEE Transactions on Software Engineering.

Key Institutional Academic Policies

Students should note the following are key policies as outlined in the University Catalogue and Students Handbook

1. *Academic dishonesty*

- a. Any intentional giving or use of external assistance during an examination without the express permission of the faculty member giving the examination.
- b. **Fabrication:** any falsification or invention of data, citation or other authority in an academic exercise;
- c. **Plagiarism:** any passing off of another's ideas, words, or work as one's own;

- d. **Previously Submitted Work:** presenting work prepared for and submitted to another course;

2. Class Attendance

Students are expected to attend all classes. Upon being absent from **five** classes in a 3 unit course, the instructor will give a student an **"F"** grade for that course.